

PHYSICS CLASS 9

COMPLETE GUESS PAPER SOLUTIONS

FBISE SSC-I | 2nd Annual 2026 | Fakhar STEM Sphere

42 short answers + 12 worked numericals + 12 long answers

Solutions match S1-S42, N1-N12 and L1-L12 in the theory guess paper. Answers use simple English, essential scientific points, clear working and labelled diagrams.

How to use: Learn the reason, not only the wording. In the exam, answer the exact question asked. For numericals show data, formula, substitution, answer and unit. Draw diagrams with labels and arrows. Marks depend on the actual board question and marking scheme; these are model practice answers.

Numerical convention: $g = 9.8 \text{ m/s}^2$, as stated in the guess paper. If an examination paper gives a different value, use that value. Round only the final result.

A. SHORT-QUESTION SOLUTIONS | S1-S42

S1. Differentiate between base and derived physical quantities. Give one example of each and write its SI unit.

Theory SLOs: P-09-A-03, P-09-A-04

Base quantities are independent physical quantities that are not expressed in terms of other quantities. Example: length, SI unit metre (m). **Derived quantities** are obtained by combining base quantities. Example: speed = distance/time, SI unit m/s. Other examples are area (m^2) and force ($\text{kg m/s}^2 = \text{N}$).

S2. Give three differences between scalar and vector quantities. Explain why displacement is a vector.

Theory SLOs: P-09-A-07, P-09-A-09

Three differences:

1. A scalar has magnitude only; a vector has magnitude and direction.
2. Scalars are added by ordinary arithmetic; vectors are added while taking direction into account.
3. Examples of scalars: mass and speed. Examples of vectors: force and displacement.

Displacement is a vector because it specifies the straight-line change in position from the starting point to the final point, together with its direction. For example, 5 m east is different from 5 m west.

S3. Explain scientific notation with an example. Show that the prefix micro is one thousand times smaller than milli.

Theory SLOs: P-09-A-05, P-09-A-06

Scientific notation: A number is written as $a \times 10^n$, where $1 \leq a < 10$ and n is an integer. Example: $45,000 \text{ m} = 4.5 \times 10^4 \text{ m}$.

Micro (μ) = 10^{-6} ; milli (m) = 10^{-3} .

Micro/milli = $10^{-6}/10^{-3} = 10^{-3} = 1/1000$.

Thus, one micrometre is one thousandth of one millimetre.

S4. A balance gives slightly different readings for the same object on repeated measurements. Identify the type of error and suggest two ways to reduce it.

Theory SLOs: P-09-A-17

The varying readings indicate a **random error**, assuming the object has not changed. Small fluctuations can arise from vibration, drafts or unstable electronic readings.

Reduce it: (1) Take several readings under the same conditions and calculate their mean. (2) Place the balance on a stable, level surface away from vibrations and drafts, and wait for the reading to settle. Averaging reduces random variation but does not correct a systematic calibration error.

S5. Differentiate between accuracy and precision. Can readings be precise but inaccurate? Give an example.

Theory SLOs: P-09-A-18

Accuracy: closeness of a measurement to the accepted or true value.

Precision: closeness of repeated readings to one another.

Yes. If the true mass is 100.0 g but readings are 105.1 g, 105.0 g and 105.1 g, the readings are very close together, so they are precise. They are far from 100.0 g, so they are inaccurate, possibly because the balance is incorrectly calibrated.

S6. Differentiate between physical and non-physical quantities with examples. Why must a measurement include both magnitude and unit?

Theory SLOs: P-09-A-01, P-09-A-02

A **physical quantity** can be measured and expressed by a numerical magnitude and a unit, for example 2 kg of mass or 5 s of time. A **non-physical quantity**, such as kindness or beauty, is not measured as a standard physical quantity with an SI unit.

A unit identifies the measurement standard. The number 5 alone is incomplete: 5 m, 5 cm and 5 km describe different lengths.

S7. Distinguish between translatory, rotatory and vibratory motion. Give one example of each.

Theory SLOs: P-09-B-01

Translatory motion: a body changes position without rotating as a whole; its points move through the same displacement at a given time. Example: a lift moving vertically.

Rotatory motion: a body turns about an axis. Example: a ceiling fan.

Vibratory motion: repeated to-and-fro motion about an equilibrium position. Example: a pendulum bob or a vibrating tuning fork.

S8. Differentiate between speed and velocity, and between average and instantaneous speed.

Theory SLOs: P-09-B-02, P-09-B-05

Speed is distance travelled per unit time and is a scalar. **Velocity** is displacement per unit time and is a vector.

Average speed = total distance/total time over an interval. **Instantaneous speed** is speed at one particular instant, such as the speedometer reading at that instant.

S9. Define acceleration and derive its SI unit. Distinguish between uniform velocity and uniform acceleration.

Theory SLOs: P-09-B-06, P-09-B-07, P-09-B-08

Acceleration is the rate of change of velocity: $a = (v - u)/t$. Its unit is $(\text{m/s})/\text{s} = \text{m/s}^2$.

Uniform velocity: equal displacements in equal time intervals in a fixed direction; acceleration is zero.

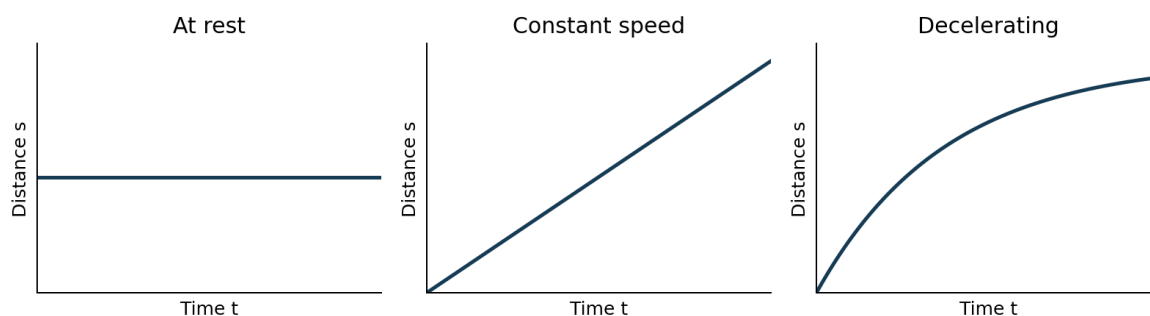
Uniform acceleration: equal changes of velocity in equal time intervals; velocity changes at a constant rate.

S10. Sketch distance-time graphs for a body at rest, moving at constant speed, and decelerating. Explain what the gradient shows.

Theory SLOs: P-09-B-09, P-09-B-11

The graphs are shown below. At rest, distance remains constant, so the graph is horizontal. At constant speed, distance increases linearly, so the graph is a straight sloping line. During deceleration while still moving forward, distance increases more slowly, so the curve gradually becomes less steep.

Gradient = change in distance/change in time = speed. A steeper distance-time graph means greater speed. A horizontal section means zero speed.



S11. What do the gradient and the area under a speed-time graph represent? Explain using their units.

Theory SLOs: P-09-B-12, P-09-B-14

Gradient: change in speed/time = acceleration for straight-line motion in a fixed direction. Unit: $(\text{m/s})/\text{s} = \text{m/s}^2$.

Area under the graph: speed $\times$ time = distance travelled. Unit: $(\text{m/s}) \times \text{s} = \text{m}$. For changing speed, add the areas of rectangles, triangles or trapezia.

S12. State Newton's first law. Use inertia to explain why passengers move forward when a moving bus stops suddenly.

Theory SLOs: P-09-B-18, P-09-B-27

Newton's first law: A body remains at rest or moves with constant velocity in a straight line unless an unbalanced external force acts on it.

When a bus stops suddenly, the passengers' feet slow with the bus but the upper part of their bodies tends to continue moving because of inertia. Therefore, passengers lean or fall forward. A seat belt provides the force needed to slow the body safely.

S13. Differentiate between mass and weight. Define gravitational field strength and write its SI unit.

Theory SLOs: P-09-B-19, P-09-B-20

Mass: amount of matter and a measure of inertia; scalar; SI unit kg; it does not change when an object is taken to another planet.

Weight: gravitational force on a mass; vector directed toward the attracting body; SI unit N; $W = mg$, so it changes when g changes.

Gravitational field strength is gravitational force per unit mass: $g = W/m$. Its SI unit is **N/kg**, equivalent to m/s^2 .

S14. Name the three fundamental interactions listed in the supplied framework: gravitational, strong and electroweak. State one feature of each and name the Pakistani Nobel laureate associated with electroweak unification.

Theory SLOs: P-09-B-25

As grouped in the supplied framework:

1. **Gravitational interaction:** attraction between masses; weakest and long-range.
2. **Strong interaction:** binds quarks and is responsible for binding protons and neutrons in nuclei; very strong over extremely short distances.
3. **Electroweak interaction:** unified description of electromagnetic and weak interactions; the weak interaction is involved in beta decay, while the electromagnetic interaction acts between electric charges.

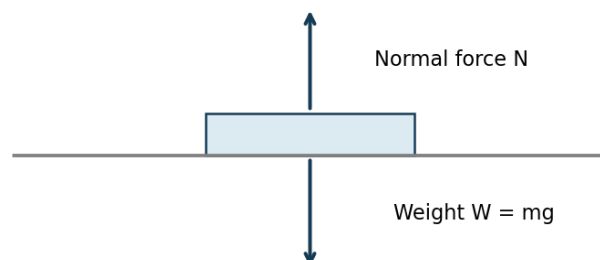
Dr Abdus Salam shared the 1979 Nobel Prize in Physics for work on electroweak unification. **Exam note:** Many texts list four fundamental interactions separately; this framework combines electromagnetic and weak interactions under electroweak.

S15. Differentiate between contact and non-contact forces. Draw a labelled free-body diagram of a book resting on a horizontal table.

Theory SLOs: P-09-B-23, P-09-B-24, P-09-B-26

Contact forces require physical contact between interacting bodies, for example friction and normal contact force. **Non-contact forces** act without direct contact, for example gravity and magnetic force.

For a book resting on a horizontal table, draw an upward normal force N and a downward weight $W = mg$. Since the book is at rest and no other vertical forces act, $N = W$ and the resultant force is zero.



S16. State Newton's third law and give two examples. Why do action and reaction forces not cancel each other on a single body?

Theory SLOs: P-09-B-31, P-09-B-32

Newton's third law: When body A exerts a force on body B, body B exerts an equal and opposite force on body A.

Examples: (1) A swimmer pushes water backward; water pushes the swimmer forward. (2) A person pushes the ground backward while walking; the ground pushes the person forward. The forces do not cancel on one body because **they act on different bodies**.

S17. State Newton's second law in terms of acceleration. Mention two situations in which Newtonian mechanics is not an adequate description.

Theory SLOs: P-09-B-30, P-09-B-33

Newton's second law: The acceleration of a body is directly proportional to the resultant force and inversely proportional to its mass. It acts in the direction of the resultant force. For constant mass, **$F = ma$** .

Newtonian mechanics is not an adequate description (1) when speed is close to the speed of light, where relativity is needed, and (2) for atomic/subatomic phenomena where quantum effects are important.

S18. Explain how friction dissipates energy. Distinguish rolling from sliding friction and state one method of reducing friction.

Theory SLOs: P-09-B-35, P-09-B-37, P-09-B-38

Friction transfers useful mechanical energy into internal energy of the rubbing surfaces and surroundings, raising their temperature. For example, rubbing hands produces warmth. The energy is **dissipated, not destroyed**.

In sliding friction, surfaces slide over one another. In rolling friction, a wheel or ball rolls over a surface; rolling resistance is usually smaller than sliding friction. Lubricating moving machine parts reduces direct surface contact and friction.

S19. Explain how a falling object reaches terminal velocity. What are its resultant force and acceleration at terminal velocity?

Theory SLOs: P-09-B-36

Initially, the object has a small speed, so air resistance is small and weight exceeds air resistance. The object accelerates downward. As speed increases, air resistance increases and resultant downward force decreases.

Eventually, upward air resistance equals downward weight. The resultant force is **zero** and acceleration is **zero**. The object continues falling at a constant speed called **terminal velocity**.

S20. Define momentum and impulse with their SI units. Explain why increasing the stopping time reduces the average force in a collision.

Theory SLOs: P-09-B-39, P-09-B-40, P-09-B-42

Momentum: product of mass and velocity, $p = mv$; SI unit kg m/s.

Impulse: force x duration of force = change in momentum, $J = F\Delta t = \Delta p$; SI unit N s, equivalent to kg m/s.

For the same change in momentum, $F_{\text{average}} = \Delta p / \Delta t$. Increasing stopping time reduces the average force. This is how airbags help reduce injury.

S21. State the principle of conservation of momentum. Explain the recoil of a gun using this principle.

Theory SLOs: P-09-B-41

Conservation of momentum: The total momentum of a system remains constant if the net external force on the system is zero.

Before firing, the gun and bullet are at rest, so total momentum is zero. After firing, the bullet gains forward momentum and the gun gains equal backward momentum. Thus $m_b v_b + m_g v_g = 0$. The gun recoils backward.

S22. Differentiate between like and unlike parallel forces. Define the moment of a force and explain why a door handle is placed far from its hinges.

Theory SLOs: P-09-B-43, P-09-B-44

Like parallel forces act along parallel lines in the same direction. **Unlike parallel forces** act along parallel lines in opposite directions.

The **moment of a force** is its turning effect about a pivot: moment = force $\times$ perpendicular distance from the pivot, $\tau = Fd$. Unit: N m. A door handle far from the hinge gives a larger perpendicular distance, so a smaller force produces the required turning effect.

S23. State the two conditions for complete equilibrium of a rigid body. Distinguish stable, unstable and neutral equilibrium using examples.

Theory SLOs: P-09-B-34, P-09-B-45

Conditions for complete equilibrium: (1) Resultant force = 0. (2) Resultant moment about a point = 0; total clockwise moment equals total anticlockwise moment.

Stable: after a small displacement the body tends to return, e.g. a ball at the bottom of a bowl.

Unstable: after a small displacement it moves farther away, e.g. a ball on top of a dome.

Neutral: it remains at its new position, e.g. a ball on a level horizontal surface.

S24. Define centre of gravity. Explain why a racing car is more stable when it has a low centre of gravity and a wide base.

Theory SLOs: P-09-B-47, P-09-B-49, P-09-B-50

The **centre of gravity** is the point through which the resultant weight of a body acts.

An object is less likely to topple while the vertical line through its centre of gravity falls inside its base. A low centre of gravity requires a greater tilt before this line moves outside the base. A wider base also increases the tilt needed. Hence a low, wide racing car is more stable on bends.

S25. Define centripetal force and state its direction. Identify the force that keeps the Moon in orbit and the force that keeps a stone moving in a circle on a string.

Theory SLOs: P-09-B-53, P-09-B-54

Centripetal force is the resultant force directed toward the centre of a circular path that changes the direction of velocity and keeps an object moving in the circle.

For the Moon, **Earth's gravitational attraction** supplies centripetal force. For a stone whirled on a string, **tension in the string** supplies it. The force points inward; the instantaneous velocity is tangential.

S26. Define spring constant and give its SI unit. Explain how Hooke's law is used in a spring balance.

Theory SLOs: P-09-B-56, P-09-B-59

The **spring constant** is force per unit extension: $k = F/x$. Unit: N/m. A larger k means a stiffer spring.

Within the proportional range, Hooke's law gives $F = kx$. In a spring balance, the weight of an object stretches the spring. Since extension is proportional to force, the scale is calibrated to show the applied force.

S27. Define work done and write its formula and SI unit. Explain why holding a stationary bag involves no mechanical work on the bag.

Theory SLOs: P-09-B-60, P-09-B-61

Work done is energy transferred when a force moves an object through a distance in the direction of the force: $W = Fd$. Unit: joule (J), where $1 \text{ J} = 1 \text{ N m}$.

When a bag is held stationary, its displacement is zero. Therefore, mechanical work on the bag is $W = F \times 0 = 0 \text{ J}$. A person may still use chemical energy internally while holding it.

S28. State the law of conservation of energy. Explain why a machine cannot continuously supply useful energy without an energy input.

Theory SLOs: P-09-B-68, P-09-B-69

Conservation of energy: Energy cannot be created or destroyed; it can only be transferred or transformed from one form to another.

A machine needs an energy source to produce useful energy output. It cannot create energy continuously from nothing. Real machines also transfer some energy to the surroundings as heat or sound, so their useful output is less than their input.

S29. Define power and efficiency. Write the formulas for power and percentage efficiency, identifying the quantities used.

Theory SLOs: P-09-B-73, P-09-B-74

Power is the rate of doing work or transferring energy: $P = W/t = E/t$. P is power in watts (W); W or E is work/energy in joules (J); t is time in seconds (s).

Efficiency is the fraction of input energy or power converted to useful output.

Efficiency (%) = (useful output energy/total input energy) $\times 100$

or (useful output power/total input power) $\times 100$. It has no unit.

S30. Differentiate renewable and non-renewable energy sources. Give two examples of each and one environmental disadvantage of fossil fuels.

Theory SLOs: P-09-B-70, P-09-B-72

Renewable sources are naturally replenished on human timescales, for example sunlight and wind.

Non-renewable sources have limited reserves that are not replaced quickly, for example coal and natural gas.

Burning fossil fuels releases carbon dioxide, contributing to climate change. It may also produce air pollutants, depending on the fuel and combustion conditions.

S31. Define pressure with its SI unit. Explain why a sharp knife cuts better than a blunt knife and why broad tyres reduce sinking in soft ground.

Theory SLOs: P-09-B-77, P-09-B-78

Pressure is normal force per unit area: $p = F/A$. SI unit: pascal (Pa), where $1 \text{ Pa} = 1 \text{ N/m}^2$.

A sharp knife has a smaller contact area, so the same force gives greater pressure and it cuts more easily. Broad tyres spread the vehicle's weight over a larger area, reducing pressure on soft ground and making sinking less likely.

S32. Explain how a drinking straw demonstrates atmospheric pressure. Would an ordinary open drinking straw work on the Moon? Justify your answer.

Theory SLOs: P-09-B-80, P-09-B-81

When air is drawn from a straw, pressure inside the straw falls. Atmospheric pressure acting on the drink's surface pushes the liquid up the straw into the mouth. This shows that the atmosphere exerts pressure.

An ordinary open drinking straw would **not work in this way on the Moon** because there is no substantial atmosphere to push the liquid upward. Creating low pressure in the straw alone is not sufficient.

S33. Differentiate between a barometer and a manometer. How can a rapid fall in atmospheric pressure help indicate a change in weather?

Theory SLOs: P-09-B-82, P-09-B-83, P-09-B-85

A **barometer** measures atmospheric pressure. A **manometer** measures the pressure difference between two points, often a gas pressure relative to atmospheric pressure.

A rapid fall in barometric pressure can indicate an approaching low-pressure weather system, often associated with cloud, wind or rain. It is an indication, not a certain weather prediction by itself.

S34. Explain why liquid pressure increases with depth. State Pascal's law and name one hydraulic application.

Theory SLOs: P-09-B-84, P-09-B-86

At a greater depth, a larger weight of liquid lies above the point. Therefore the force per unit area is greater: $p = \rho gh$ for gauge pressure in a liquid of uniform density.

Pascal's law: A pressure change applied to a confined fluid is transmitted undiminished throughout the fluid and to the container walls. A hydraulic lift or hydraulic brake system uses this principle.

S35. Define density and give its formula and SI unit. Explain why equal volumes of two substances can have different masses.

Theory SLOs: P-09-C-01

Density is mass per unit volume: $\rho = m/V$. SI unit: kg/m^3 .

For equal volumes, $m = \rho V$. A substance with greater density has greater mass in the same volume, because its particles have different masses and/or are packed more closely.

S36. Explain how particle motion changes as temperature rises. Relate this change to the internal energy of a substance.

Theory SLOs: P-09-C-03, P-09-C-05, P-09-C-06

When temperature rises, particles have greater average kinetic energy: particles in a solid vibrate more vigorously, while particles in a liquid or gas generally move faster.

Internal energy is the total microscopic kinetic and potential energy of the particles. For a substance remaining in the same state, a temperature rise increases the kinetic-energy part and hence its internal energy.

S37. What is plasma? Give two examples. State the approximate Celsius temperature of absolute zero and explain its meaning.

Theory SLOs: P-09-C-04, P-09-C-05

Plasma is an ionized state of matter containing free electrons and ions. Examples: matter in stars and the glowing gas in a neon discharge lamp.

Absolute zero is approximately $-273\text{ }^{\circ}\text{C}$ (0 K), the lowest possible temperature. Particles have their minimum possible kinetic energy. Do not describe this as complete absence of all quantum motion.

S38. What is a thermometric property? Why are fixed points needed for thermometer calibration? Explain how a narrow capillary affects sensitivity.

Theory SLOs: P-09-C-07, P-09-C-08, P-09-C-11

A **thermometric property** is a physical property that changes predictably with temperature, for example the volume of mercury or alcohol, or a thermocouple's voltage.

Fixed points provide reproducible reference temperatures for calibration. At standard atmospheric pressure, the ice point is $0\text{ }^{\circ}\text{C}$ and the steam point is $100\text{ }^{\circ}\text{C}$. For the same expansion of liquid, a narrower capillary produces a larger rise in length, so it increases sensitivity.

S39. Define induced magnetism. Give two differences between a permanent magnet and an electromagnet and one use of each.

Theory SLOs: P-09-E-02, P-09-E-04, P-09-E-10

Induced magnetism occurs when a magnetic material becomes magnetized due to an external magnetic field.

Permanent magnet: retains magnetism without an electric current; its magnetism cannot usually be switched off easily. Use: a compass.

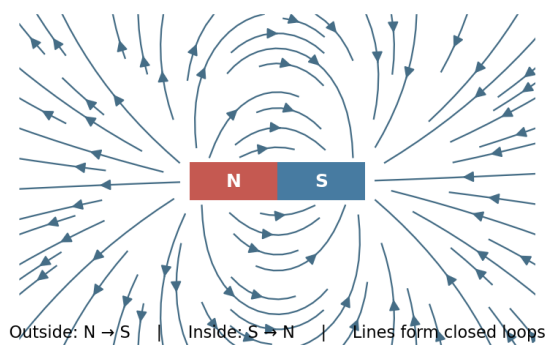
Electromagnet: magnetism is produced by current in a coil, often with a soft-iron core; it can be switched on/off and adjusted by changing current. Use: an electric bell or a scrap-metal lifting crane.

S40. Draw the magnetic field around a bar magnet, showing direction. Explain how the spacing of field lines indicates field strength.

Theory SLOs: P-09-E-05, P-09-E-07, P-09-E-08, P-09-E-09

The field pattern is shown below. **Outside the magnet**, field lines run from N to S. **Inside**, they return from S to N, forming closed loops. At a point, field direction is the direction in which a north test pole would be pushed.

Closely spaced lines indicate a **stronger** field; widely spaced lines indicate a weaker field. The field is strongest near the poles. Field lines do not cross.



S41. Compare the responses of paramagnetic and diamagnetic materials to an external magnetic field. Explain why soft iron is used for magnetic shielding.

Theory SLOs: P-09-E-12, P-09-E-15

Paramagnetic materials: are weakly attracted by an external magnetic field; their induced magnetization is in the direction of the field. Example: aluminium.

Diamagnetic materials: are weakly repelled; their induced magnetization is opposite to the field. Example: bismuth.

Soft iron is a soft ferromagnetic material with high magnetic permeability. It provides an easy path for magnetic flux, diverting much of the external field around a protected region. This reduces the field inside a suitable soft-iron enclosure.

S42. Briefly describe astronomy, particle physics and geophysics. Explain what is meant by a scientific theory and how evidence is used to develop it.

Theory SLOs: P-09-G-02, P-09-G-06

Astronomy: study of celestial objects and their motions.

Particle physics: study of fundamental particles and their interactions.

Geophysics: use of physics to study Earth, such as earthquakes and its magnetic field.

A **scientific theory** is a well-supported explanation of observations. Scientists propose explanations, make testable predictions and compare them with repeated observations and experiments. A theory is refined or replaced if reliable evidence conflicts with its predictions. A theory does not become a law simply by collecting more evidence.

B. WORKED NUMERICALS | N1-N2

N1. (a) Convert 25 micrometres to kilometres. (b) Express 0.0000034 s using a suitable SI prefix. (c) Write 72,000,000 N in scientific notation.

Theory SLOs: P-09-A-05, P-09-A-06

(a) Micrometres to kilometres

$$1 \mu\text{m} = 10^{-6} \text{ m and } 1 \text{ m} = 10^{-3} \text{ km.}$$

$$25 \mu\text{m} = 25 \times 10^{-6} \times 10^{-3} \text{ km}$$

$$= 2.5 \times 10^{-8} \text{ km.}$$

(b) Suitable prefix

$$0.0000034 \text{ s} = 3.4 \times 10^{-6} \text{ s} = 3.4 \mu\text{s.}$$

(c) Scientific notation

$$72,000,000 \text{ N} = 7.2 \times 10^7 \text{ N.}$$

Check: Moving from micrometres to kilometres makes the numerical value very small. Do not confuse micro (10^{-6}) with milli (10^{-3}).

N2. Two perpendicular displacements are 6 m east and 8 m north. Calculate the magnitude of their resultant and its angle measured north of east. Draw a labelled vector diagram.

Theory SLOs: P-09-A-10

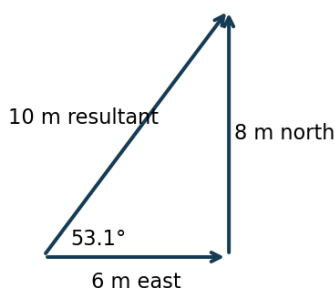
Given: eastward displacement = 6 m; northward displacement = 8 m. The two directions are perpendicular.

Magnitude: $R = \sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ m.}$

Direction: $\tan \theta = \text{northward/eastward} = 8/6.$

$$\theta = \tan^{-1}(8/6) = 53.1^\circ \text{ north of east.}$$

Final displacement: 10 m at 53.1° north of east. The equivalent direction is 36.9° east of north. Use the labelled head-to-tail diagram; the resultant joins the starting point to the final point.



B. WORKED NUMERICALS | N3-N4

N3. A car travels 6 m in 2 s, then 8 m in 4 s, then 10 m in 2 s. Calculate (a) total distance, (b) total time and (c) average speed. Explain why the mean of the three individual speeds is not appropriate here.

Theory SLOs: P-09-B-03, P-09-B-04

Given: distances 6 m, 8 m, 10 m; times 2 s, 4 s, 2 s.

(a) Total distance: $s = 6 + 8 + 10 = 24 \text{ m}$.

(b) Total time: $t = 2 + 4 + 2 = 8 \text{ s}$.

(c) Average speed: $v_{\text{avg}} = \text{total distance} / \text{total time} = 24 / 8 = 3.0 \text{ m/s}$.

The individual speeds are 3, 2 and 5 m/s. Their arithmetic mean is $(3 + 2 + 5) / 3 = 3.33 \text{ m/s}$, which is not the journey's average speed because the time intervals are unequal. The correct time-weighted average is $(3 \times 2 + 2 \times 4 + 5 \times 2) / 8 = 3.0 \text{ m/s}$.

N4. A car accelerates uniformly from rest to 20 m/s in 5 s, continues at 20 m/s for 10 s, and then slows uniformly to rest in 5 s. Draw its speed-time graph. Calculate (a) initial acceleration, (b) acceleration while stopping and (c) total distance from the graph.

Theory SLOs: P-09-B-07, P-09-B-12, P-09-B-14

Graph: join the points (0 s, 0 m/s), (5 s, 20 m/s), (15 s, 20 m/s) and (20 s, 0 m/s) with straight lines.

(a) Initial acceleration: $a = (20 - 0) / 5 = 4.0 \text{ m/s}^2$.

(b) Acceleration while stopping: $a = (0 - 20) / (20 - 15) = -4.0 \text{ m/s}^2$. The magnitude of deceleration is 4.0 m/s^2 .

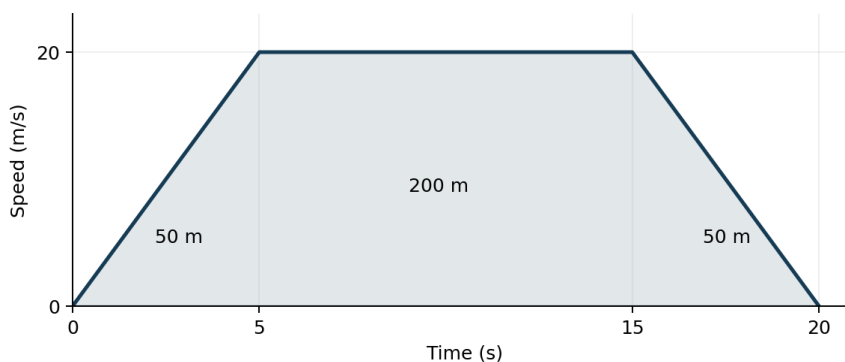
(c) Distance = total area under the speed-time graph.

First triangle: $(1/2) \times 5 \times 20 = 50 \text{ m}$.

Rectangle: $10 \times 20 = 200 \text{ m}$.

Last triangle: $(1/2) \times 5 \times 20 = 50 \text{ m}$.

Total distance = $50 + 200 + 50 = 300 \text{ m}$.



B. WORKED NUMERICALS | N5-N6

N5. A stone is dropped from rest and reaches the ground at 29.4 m/s. Ignore air resistance. Calculate (a) time of fall and (b) height of release, using the area under its speed-time graph.

Theory SLOs: P-09-B-10, P-09-B-12

Given: $u = 0$; $v = 29.4$ m/s downward; $g = 9.8$ m/s²; air resistance ignored.

(a) Time of fall:

$$g = (v - u)/t, \text{ so } t = (v - u)/g = 29.4/9.8 = \mathbf{3.0 \text{ s.}}$$

(b) Height: The speed-time graph is a straight line from (0, 0) to (3 s, 29.4 m/s).

Height = triangular area = $(1/2) \times \text{time} \times \text{final speed}$

$$h = (1/2) \times 3.0 \times 29.4 = \mathbf{44.1 \text{ m.}}$$

Check: Average speed during uniform acceleration from rest is 14.7 m/s. Hence $h = 14.7 \times 3.0 = 44.1$ m. Do not use the final speed as the average speed.

N6. A 2.0 kg object changes velocity from 3.0 m/s to 7.0 m/s in the same direction in 0.20 s. Calculate (a) change in momentum, (b) impulse and (c) average resultant force.

Theory SLOs: P-09-B-39, P-09-B-40, P-09-B-42

Given: $m = 2.0$ kg; $u = 3.0$ m/s; $v = 7.0$ m/s; $\Delta t = 0.20$ s. Take the direction of motion as positive.

(a) Change in momentum:

$$\Delta p = m(v - u) = 2.0(7.0 - 3.0) = \mathbf{8.0 \text{ kg m/s.}}$$

(b) Impulse: $J = \Delta p = \mathbf{8.0 \text{ N s}}$ in the direction of motion.

(c) Average resultant force:

$$F = \Delta p / \Delta t = 8.0 / 0.20 = \mathbf{40 \text{ N}}$$
 in the direction of motion.

Check: $a = 4.0 / 0.20 = 20$ m/s²; $ma = 2.0 \times 20 = 40$ N.

B. WORKED NUMERICALS | N7-N8

N7. A 2.0 kg trolley moving at 4.0 m/s strikes a stationary 3.0 kg trolley. They stick together. Neglect external horizontal forces. Calculate their common velocity and check that total momentum is conserved.

Theory SLOs: P-09-B-41

Given: $m_1 = 2.0 \text{ kg}$; $u_1 = 4.0 \text{ m/s}$; $m_2 = 3.0 \text{ kg}$; $u_2 = 0$. The trolleys stick together, so both have final velocity v .

Conservation of momentum:

$$m_1 u_1 + m_2 u_2 = (m_1 + m_2) v$$

$$2.0 \times 4.0 + 3.0 \times 0 = (2.0 + 3.0) v$$

$8.0 = 5.0 v$, so **$v = 1.6 \text{ m/s}$** in the original direction of the moving trolley.

Check: Initial momentum = 8.0 kg m/s .

Final momentum = $5.0 \times 1.6 = 8.0 \text{ kg m/s}$.

Thus total momentum is conserved. Kinetic energy need not be conserved in a collision where the trolleys stick together.

N8. A light horizontal beam is supported at its centre. A 30 N downward force acts 0.40 m to the left of the pivot. What downward force applied 0.60 m to the right balances the beam? Calculate each moment and the upward support force.

Theory SLOs: P-09-B-44, P-09-B-45

Given: left force = 30 N at 0.40 m ; right force = F at 0.60 m . The beam is light, so its weight is neglected.

Principle of moments: anticlockwise moment = clockwise moment.

$$30 \times 0.40 = F \times 0.60$$

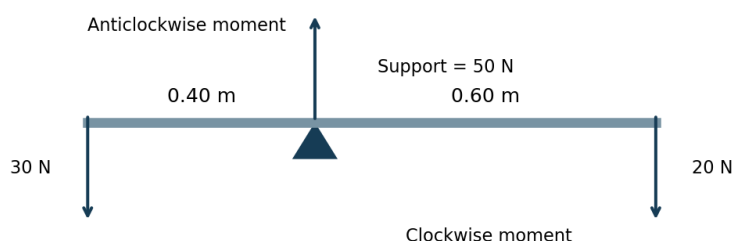
$$F = 12/0.60 = \mathbf{20 \text{ N downward.}}$$

Moments: left = $30 \times 0.40 = \mathbf{12 \text{ N m anticlockwise.}}$

Right = $20 \times 0.60 = \mathbf{12 \text{ N m clockwise.}}$

Vertical equilibrium: upward support force = total downward force.

$R = 30 + 20 = \mathbf{50 \text{ N upward.}}$ The resultant force and resultant moment are both zero.



B. WORKED NUMERICALS | N9-N10

N9. An ideal spring extends by 4.0 cm under a force of 12 N. Calculate (a) its spring constant and (b) the force required for a 7.0 cm extension, assuming force remains proportional to extension.

Theory SLOs: P-09-B-56

Given: $F = 12 \text{ N}$; $x = 4.0 \text{ cm} = 0.040 \text{ m}$.

(a) Spring constant:

$$k = F/x = 12/0.040 = \mathbf{300 \text{ N/m}}.$$

(b) Force for 7.0 cm extension:

$$x = 7.0 \text{ cm} = 0.070 \text{ m}.$$

$$F = kx = 300 \times 0.070 = \mathbf{21 \text{ N}}.$$

Check: The extension rises by a factor $7/4 = 1.75$, so the force becomes $12 \times 1.75 = 21 \text{ N}$. The extension must be in metres when k is in N/m .

N10. A football of mass 0.45 kg is kicked vertically upward at 22 m/s. Ignore air resistance in part (a). Calculate (a) its maximum height using energy conservation. If it actually rises to only 20 m, calculate (b) work done against air resistance on the upward journey.

Theory SLOs: P-09-B-61, P-09-B-64, P-09-B-65, P-09-B-66

Given: $m = 0.45 \text{ kg}$; $u = 22 \text{ m/s}$; $g = 9.8 \text{ m/s}^2$. At maximum height, speed is zero.

(a) Without air resistance:

Initial kinetic energy = final gravitational potential energy.

$$(1/2)mu^2 = mgh; \text{ cancel } m.$$

$$h = u^2/(2g) = 22^2/(2 \times 9.8)$$

$$h = 484/19.6 = \mathbf{24.7 \text{ m}} \text{ (approximately).}$$

(b) Actual height = 20 m:

$$\text{Initial KE} = (1/2) \times 0.45 \times 22^2 = 108.9 \text{ J}.$$

$$\text{Final PE} = 0.45 \times 9.8 \times 20 = 88.2 \text{ J}.$$

$$\text{Energy transferred against resistance} = 108.9 - 88.2 = \mathbf{20.7 \text{ J}}.$$

Sign: Work done against air resistance is $+20.7 \text{ J}$. Work done by air resistance on the ball is -20.7 J . The final KE is zero at the highest point.

B. WORKED NUMERICALS | N11-N12

N11. A hydraulic lift supports a car of mass 2500 kg. The small and large piston areas are 6.0 cm^2 and 90 cm^2 . Neglect piston weight and friction. Calculate (a) the car's weight, (b) the required force on the small piston and (c) the fluid pressure.

Theory SLOs: P-09-B-19, P-09-B-86

Given: $m = 2500 \text{ kg}$; $A_1 = 6.0 \text{ cm}^2 = 6.0 \times 10^{-4} \text{ m}^2$; $A_2 = 90 \text{ cm}^2 = 9.0 \times 10^{-3} \text{ m}^2$.

(a) Weight: $F_2 = mg = 2500 \times 9.8 = \mathbf{24,500 \text{ N}}$.

(b) Force on small piston:

$$\begin{aligned} F_1/A_1 &= F_2/A_2 \\ F_1 &= 24,500 \times (6.0/90) = \mathbf{1633 \text{ N} \approx 1.63 \times 10^3 \text{ N}}. \end{aligned}$$

(c) Transmitted gauge pressure:

$$\begin{aligned} p &= F_2/A_2 = 24,500/(9.0 \times 10^{-3}) \\ p &\approx \mathbf{2.72 \times 10^6 \text{ Pa}}. \end{aligned}$$

Using F_1/A_1 gives the same pressure. The piston area ratio is 15, so output force is 15 times input force.

N12. A satellite moves in a circular orbit of radius $7.0 \times 10^6 \text{ m}$ and period 6000 s. Calculate (a) the distance travelled in one orbit and (b) its average orbital speed. Convert the speed to km/s. Use $\pi = 3.142$.

Theory SLOs: P-09-F-01

Given: $r = 7.0 \times 10^6 \text{ m}$; $T = 6000 \text{ s}$; $\pi = 3.142$. The radius is measured from Earth's centre, not from its surface.

(a) Distance in one orbit:

$$\begin{aligned} s &= 2\pi r = 2 \times 3.142 \times 7.0 \times 10^6 \\ s &= \mathbf{4.3988 \times 10^7 \text{ m} \approx 4.40 \times 10^7 \text{ m}}. \end{aligned}$$

(b) Average orbital speed:

$$\begin{aligned} v &= s/T = (4.3988 \times 10^7)/6000 \\ v &\approx \mathbf{7.33 \times 10^3 \text{ m/s} = 7.33 \text{ km/s}}. \end{aligned}$$

Check: Divide m/s by 1000 to convert to km/s. For a complete circular orbit, average speed is nonzero, but average velocity over that full orbit is zero because displacement is zero.

C. LONG-QUESTION SOLUTIONS | L1

L1. Motion graphs

(a) Sketch a distance-time graph for an object initially at rest, then moving at constant speed, then accelerating and finally at rest again. (3 marks)

(b) Explain how the gradient shows speed. (1 mark)

(c) Using a simple speed-time graph, show that its gradient gives acceleration and its area gives distance travelled. (2 marks)

Theory SLOs: P-09-B-09, P-09-B-11, P-09-B-12, P-09-B-15

(a) Distance-time graph (3 marks)

Draw and label the two axes. The initial rest stage is horizontal. Constant speed is a straight rising section of constant gradient. During acceleration, the curve becomes progressively steeper. The final rest stage is horizontal at the new distance. Label each stage as shown below.

(b) Gradient represents speed (1 mark)

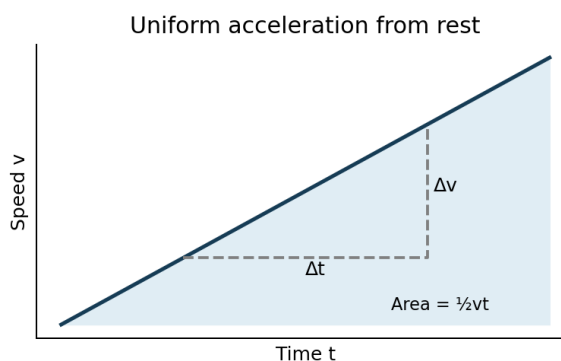
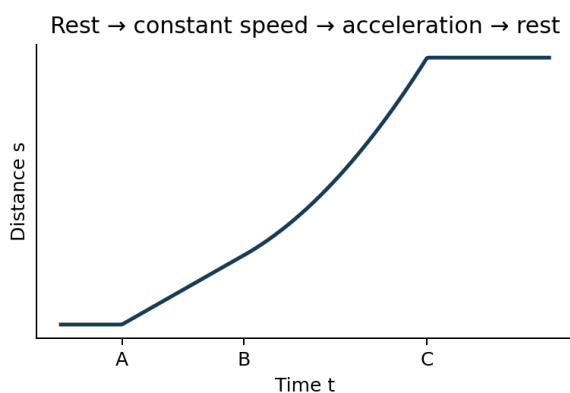
For a distance-time graph, gradient = $\Delta s / \Delta t$ = speed. A steeper section means greater speed; a horizontal section has zero gradient and therefore zero speed.

(c) Speed-time graph (2 marks)

For straight-line motion in a fixed direction, gradient = $\Delta v / \Delta t$ = acceleration. Its unit is m/s^2 .

For constant speed, the rectangular area is vt = distance. For uniform acceleration from rest, the triangular area is $(1/2)vt$ = average speed $\times$ time = distance. For a general uniformly accelerating section, the trapezium area gives the distance. Thus **distance travelled equals area under the speed-time graph**.

Exam care: On a distance-time graph, deceleration is a curve that becomes less steep while distance continues to increase. A downward slope should not be used to represent accumulated distance travelled.



C. LONG-QUESTION SOLUTIONS | L2

L2. Newton's laws in daily life

(a) State Newton's three laws of motion. (3 marks)

(b) Apply them respectively to a passenger in a suddenly stopping bus, a trolley pushed by an unbalanced force, and a swimmer moving forward. Identify the two bodies in the action-reaction pair. (3 marks)

Theory SLOs: P-09-B-27, P-09-B-30, P-09-B-31, P-09-B-32

(a) Newton's laws (3 marks)

First law: A body remains at rest or in uniform straight-line motion unless an unbalanced external force acts on it.

Second law: For constant mass, acceleration is proportional to resultant force and inversely proportional to mass; $\mathbf{F} = m\mathbf{a}$ in the direction of the resultant force.

Third law: If body A exerts a force on body B, body B exerts an equal and opposite force on body A.

(b) Applications (3 marks)

Passenger: When a bus stops suddenly, the passenger's body tends to keep moving forward because of inertia. This illustrates the first law.

Trolley: A resultant push accelerates the trolley: $a = F/m$. A greater resultant force gives greater acceleration; for the same force, a larger mass gives smaller acceleration. This illustrates the second law.

Swimmer: The swimmer pushes water backward, and the water pushes the swimmer forward. The paired forces are force of swimmer on water and force of water on swimmer. This illustrates the third law.

Important distinction: Action and reaction are equal and opposite but act on different bodies. The upward normal force and downward weight of a resting book balance on the book; they are not an action-reaction pair.

C. LONG-QUESTION SOLUTIONS | L3

L3. Friction and terminal velocity

(a) Explain how friction in machine parts transfers mechanical energy to internal energy. (2 marks)

(b) Explain two methods of reducing friction. (2 marks)

(c) Describe the changing forces and acceleration on a falling object until it reaches terminal velocity. (2 marks)

Theory SLOs: P-09-B-35, P-09-B-36, P-09-B-38

(a) Energy dissipation by friction (2 marks)

Friction opposes relative sliding between surfaces. Work must be done against this force. The useful mechanical energy becomes microscopic kinetic and potential energy of particles in the surfaces and surroundings. Machine parts become warm; some energy may also be transferred as sound. Energy is conserved, but less remains available as useful mechanical energy.

(b) Two methods to reduce friction (2 marks)

Lubrication: Oil or grease forms a layer between surfaces, reducing direct contact and sliding resistance.

Ball bearings: Balls or rollers replace much of the sliding with rolling; rolling resistance is generally smaller, so less energy is dissipated.

(c) Reaching terminal velocity (2 marks)

At release, the speed is zero and air resistance is negligible. Weight acts downward and the object accelerates downward. As speed rises, upward air resistance increases. The resultant force $W - D$ decreases, so acceleration decreases.

At terminal velocity, **$D = W$, resultant force = 0, and acceleration = 0**. The object continues moving downward at constant speed.

Diagram to draw: For an object below terminal speed, show a longer downward weight arrow and a shorter upward drag arrow. At terminal speed, draw equal opposing arrows. The weight stays essentially constant throughout.

C. LONG-QUESTION SOLUTIONS | L4

L4. Momentum, impulse and safety

(a) Define momentum and impulse and state conservation of momentum. (3 marks)

(b) Show that average resultant force equals change in momentum divided by time. (1 mark)

(c) Explain how airbags and a cricketer moving their hands backward while catching a ball reduce the average force. (2 marks)

Theory SLOs: P-09-B-39, P-09-B-40, P-09-B-41, P-09-B-42

(a) Definitions and principle (3 marks)

Momentum: $p = mv$, a vector quantity with SI unit kg m/s.

Impulse: $J = F_{\text{average}} \Delta t = \Delta p$, with SI unit N s.

Conservation of momentum: If the resultant external force on a system is zero, total momentum before an interaction equals total momentum afterward.

(b) Force in terms of momentum (1 mark)

For constant mass, $F = ma = m(v - u)/\Delta t$.

Since $m(v - u) = mv - mu = \Delta p$,

$F_{\text{average}} = \Delta p / \Delta t$. Multiplying by Δt gives impulse = change in momentum.

(c) Safety applications (2 marks)

Airbag: A passenger has to lose approximately the same momentum when stopping. The airbag increases stopping time, so the average force $\Delta p / \Delta t$ is smaller. It also spreads the force over a larger body area.

Catching a ball: A cricketer moves their hands backward while catching. The ball is brought to rest over a longer time, so the average force on the ball and hands is reduced.

Exam care: The safety device reduces the force by increasing time; it does not necessarily reduce the required change in momentum.

C. LONG-QUESTION SOLUTIONS | L5

L5. Moments and stability

- (a) Define the moment of a force and state the principle of moments. (2 marks)
(b) Draw a labelled balanced-beam diagram and explain its rotational equilibrium. (2 marks)
(c) Explain how a low centre of gravity and a broad base improve stability, using a vehicle as an example. (2 marks)

Theory SLOs: P-09-B-34, P-09-B-44, P-09-B-45, P-09-B-49, P-09-B-50

(a) Moment and principle of moments (2 marks)

The moment of a force is its turning effect about a pivot: $\tau = Fd$, where d is the perpendicular distance from the pivot to the force's line of action. Unit: N m.

For rotational equilibrium, total clockwise moment equals total anticlockwise moment about the same pivot.

(b) Balanced beam (2 marks)

The diagram below shows a light beam with 30 N acting 0.40 m to the left and 20 N acting 0.60 m to the right.

Anticlockwise moment = $30 \times 0.40 = 12 \text{ N m}$.

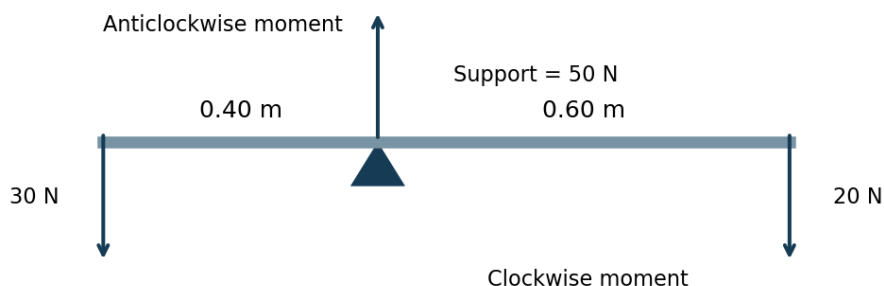
Clockwise moment = $20 \times 0.60 = 12 \text{ N m}$.

The moments balance, so there is no angular acceleration. The upward support force is 50 N, equal to the total downward force, so translational equilibrium also holds.

(c) Stability of vehicles (2 marks)

A vehicle tends to topple when the vertical line through its centre of gravity falls outside its support base. Lowering the centre of gravity and widening the base mean that a greater tilt is required for this to happen. Therefore, racing cars are designed with a low body and a wide wheel track.

Exam care: Use the perpendicular lever arm, not an arbitrary distance along a sloping force.



C. LONG-QUESTION SOLUTIONS | L6

L6. Circular motion and orbital speed

- (a) Define centripetal force and show its direction on a circular-path diagram. (2 marks)
(b) Identify its source for a stone on a string and the Moon orbiting Earth. (2 marks)
(c) Derive average orbital speed $v = 2\pi r / T$ from the distance travelled in one circular orbit. (2 marks)

Theory SLOs: P-09-B-53, P-09-B-54, P-09-F-01

(a) Centripetal force (2 marks)

Centripetal force is the inward resultant force that keeps a body moving along a circular path by continuously changing the direction of its velocity. It points toward the centre at every position. The velocity is tangential to the path, perpendicular to the radius.

(b) Sources of force (2 marks)

For a stone attached to a string, tension supplies the inward force. For the Moon orbiting Earth, Earth's gravitational attraction supplies the inward force. Centripetal force is not an additional separate interaction; it is the name for the inward resultant force.

(c) Average orbital speed (2 marks)

For an orbit of radius r , distance travelled in one revolution = circumference = $2\pi r$.

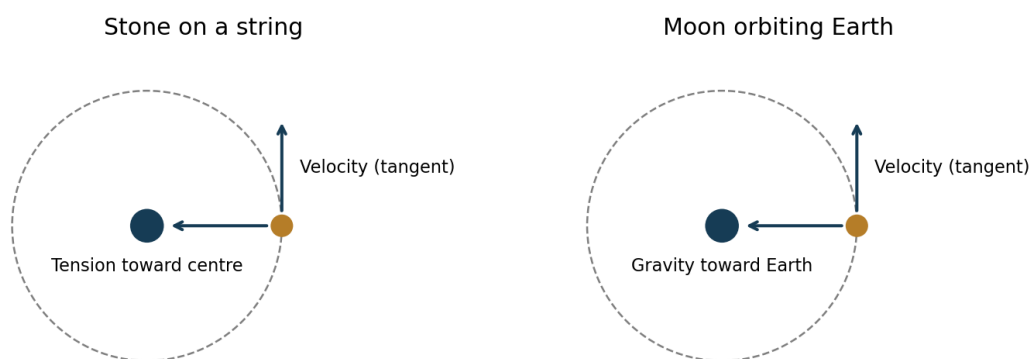
If one revolution takes period T , then

Average speed = total distance/total time

$$v = 2\pi r / T.$$

With r in metres and T in seconds, v is in m/s.

Exam care: Orbital radius is measured from the centre of the orbit. After one complete orbit the displacement is zero, so average velocity is zero even though average orbital speed is not.



C. LONG-QUESTION SOLUTIONS | L7

L7. Kinetic and gravitational potential energy

(a) Derive kinetic energy = $(1/2)mv^2$ using the triangular area under a speed-time graph for uniform acceleration from rest, together with $F = ma$ and $W = Fs$. (3 marks)

(b) Derive gravitational potential energy = mgh for lifting an object at constant speed. (2 marks)

(c) State the energy conversion during a fall without air resistance. (1 mark)

Theory SLOs: P-09-B-61, P-09-B-64, P-09-B-65, P-09-B-66

(a) Derivation of kinetic energy using a graph (3 marks)

Consider a body of mass m accelerated uniformly from rest to speed v in time t by a constant resultant force, with no dissipative loss.

Gradient of the speed-time graph: $a = v/t$.

Distance travelled = triangular area under the graph: $s = (1/2)vt$.

From Newton's second law, $F = ma = mv/t$.

Work done = $Fs = (mv/t) \times ((1/2)vt) = (1/2)mv^2$.

This work becomes kinetic energy, so $E_k = (1/2)mv^2$.

(b) Derivation of gravitational potential energy (2 marks)

Lift an object of mass m through vertical height h at constant speed near Earth. The lifting force equals its weight: $F = mg$.

Work against gravity = $Fh = mgh$.

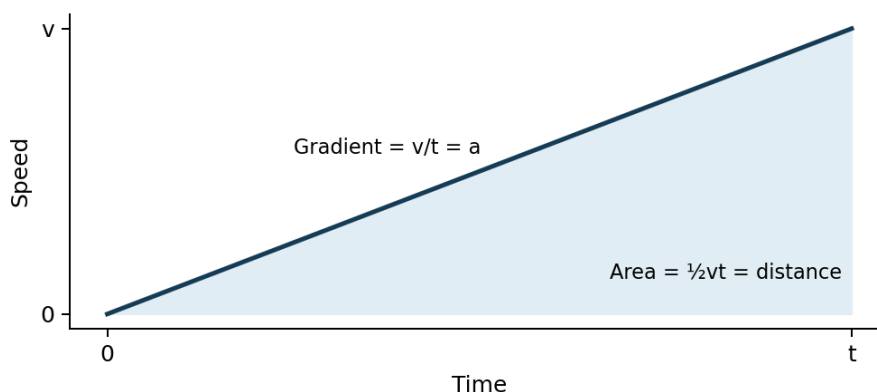
This energy is stored as gravitational potential energy, so

$\Delta E_p = mgh$. This assumes g is approximately constant.

(c) Energy conversion in free fall (1 mark)

Without air resistance, gravitational potential energy decreases and kinetic energy increases by the same amount. Hence $E_k + E_p = \text{constant}$.

Units: Both kinetic and potential energy are measured in joules (J).



C. LONG-QUESTION SOLUTIONS | L8

L8. Energy resources, power and efficiency

(a) Describe the energy transfers in a hydroelectric power station, identifying the turbine and generator. (2 marks)

(b) Compare solar and fossil-fuel electricity generation in terms of availability and environmental effects. (2 marks)

(c) Define power and percentage efficiency and write their formulas. (2 marks)

Theory SLOs: P-09-B-68, P-09-B-71, P-09-B-72, P-09-B-73, P-09-B-74

(a) Hydroelectric power station (2 marks)

Water stored at height has gravitational potential energy. As water flows downward, this energy becomes kinetic energy. The flowing water turns a **turbine**, producing rotational mechanical energy. The turbine drives a **generator**, which transfers mechanical energy into electrical energy. Some energy becomes heat and sound.

Sequence: gravitational potential → kinetic → turbine mechanical → electrical energy.

(b) Solar versus fossil-fuel electricity (2 marks)

Solar: renewable; output depends on sunshine, so it falls in cloud and stops at night without stored energy. Solar panels produce no combustion gases during operation.

Fossil fuels: non-renewable; power stations can usually generate on demand while fuel is available. Burning fuel releases carbon dioxide and may release other air pollutants. Both methods have environmental effects from construction and resource use.

(c) Power and efficiency (2 marks)

Power is work done or energy transferred per unit time: $P = W/t = E/t$. Unit: watt (W), where $1\text{ W} = 1\text{ J/s}$.

Percentage efficiency = (useful output energy/total input energy) × 100
or (useful output power/total input power) × 100.

Exam care: Energy is not lost from existence; dissipated energy is transferred to the surroundings and is less useful for the intended task.

C. LONG-QUESTION SOLUTIONS | L9

L9. Liquid pressure and hydraulic machines

(a) Derive the gauge pressure at depth h in a liquid: $p = \rho gh$, using a liquid column of cross-sectional area A . (3 marks)

(b) State Pascal's law. Draw a labelled hydraulic lift and explain how it multiplies force, using $F_1/A_1 = F_2/A_2$. (3 marks)

Theory SLOs: P-09-B-84, P-09-B-86

(a) Derivation of liquid pressure (3 marks)

Consider a vertical column of liquid of depth h , cross-sectional area A and uniform density ρ .

Volume = Ah .

Mass = density $\times$ volume = ρAh .

Weight = $mg = \rho Ahg$.

Pressure due to this column = weight/area = $\rho Ahg/A$.

$p = \rho gh$.

Thus pressure increases with depth, density and gravitational field strength. This is gauge pressure due to the liquid; if its surface is open to the atmosphere, absolute pressure is $p_{\text{atm}} + \rho gh$.

(b) Pascal's law and hydraulic lift (3 marks)

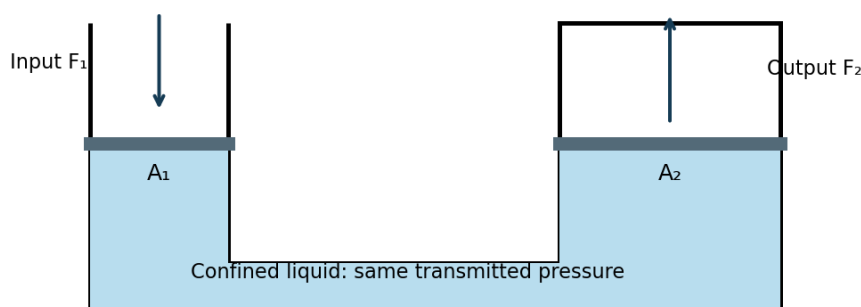
Law: A pressure change applied to a confined fluid is transmitted undiminished throughout the fluid and to the walls of its container.

Two pistons of areas A_1 and A_2 act on the same confined liquid. For ideal pistons at the same level, their transmitted pressures are equal:

$$F_1/A_1 = F_2/A_2.$$

Therefore $F_2 = F_1(A_2/A_1)$. A large output piston gives a greater force and can raise a car.

Why energy is conserved: The small piston must move a greater distance than the large piston. In an ideal system $A_1 d_1 = A_2 d_2$, giving $F_1 d_1 = F_2 d_2$. Force multiplication does not create energy.



C. LONG-QUESTION SOLUTIONS | L10

L10. Thermometers

(a) Explain the construction and working of a liquid-in-glass thermometer and a thermocouple thermometer. (4 marks)

(b) Explain how capillary width and bulb size affect the sensitivity of a liquid-in-glass thermometer, and how a fixed capillary length links sensitivity to range. (2 marks)

Theory SLOs: P-09-C-07, P-09-C-08, P-09-C-09, P-09-C-10, P-09-C-11

(a) Construction and working (4 marks)

Liquid-in-glass thermometer: A glass bulb contains mercury or coloured alcohol and is connected to a narrow capillary tube carrying a calibrated scale. When temperature rises, the liquid expands more than the glass and its column rises. When temperature falls, the liquid contracts and the column falls. The column position gives temperature after thermal equilibrium is reached.

Thermocouple: Two wires of different metals form a measuring junction. Their other ends connect to a voltage-measuring instrument with a known reference temperature. A temperature difference between the measuring and reference junctions produces a small thermoelectric voltage. The calibrated voltage is used to determine the measuring-junction temperature. Thermocouples can respond quickly and operate over a wide range, depending on their metals.

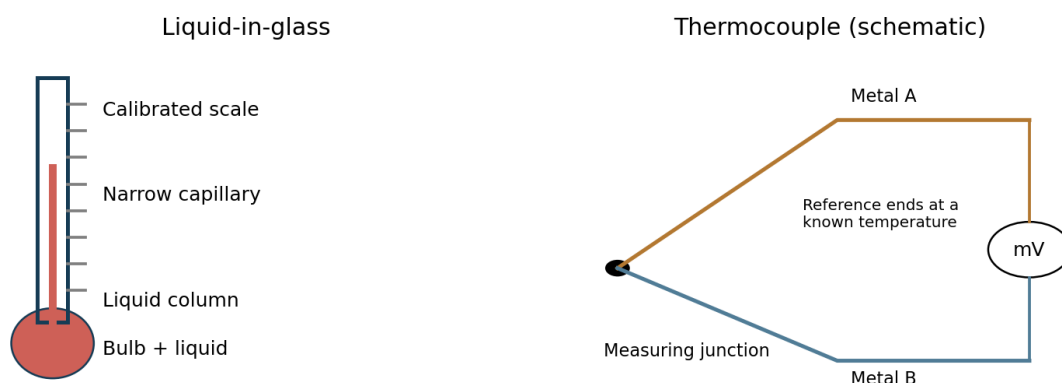
(b) Sensitivity and range (2 marks)

Narrow capillary: The same volume increase causes a greater rise in column length, increasing sensitivity.

Larger bulb: More liquid expands for the same temperature rise, giving a greater column movement and increasing sensitivity.

Fixed capillary length: A larger movement per degree uses up the available length more quickly, so a highly sensitive design generally has a smaller temperature range. A longer capillary can increase the range.

Remember: Sensitivity means change of reading per unit temperature change. Range is the interval between the lowest and highest temperatures that can be measured.



C. LONG-QUESTION SOLUTIONS | L11

L11. Magnetization and magnetic domains

- (a) Explain magnetization and demagnetization using domain theory. (2 marks)
- (b) Describe one method of magnetizing a steel bar and one method of demagnetizing it. (2 marks)
- (c) Explain induced magnetism in soft iron and draw the field pattern around a bar magnet with arrows. (2 marks)

Theory SLOs: P-09-E-02, P-09-E-04, P-09-E-07, P-09-E-11

(a) Domain explanation (2 marks)

A ferromagnetic material contains small regions called **magnetic domains**. Within each domain, magnetic moments are aligned. In an unmagnetized sample the domain directions mostly cancel, giving little net magnetism. An applied field makes favourably oriented domains grow and aligns more magnetic moments, producing net magnetization.

During demagnetization, the domain arrangement becomes less ordered so the net magnetism is reduced.

(b) One magnetizing and one demagnetizing method (2 marks)

Magnetizing: Place a steel bar inside a solenoid and pass direct current through the coil. The coil's field aligns domains. Steel can retain much of this alignment after removal of the field.

Demagnetizing: Place the magnet in a solenoid carrying alternating current, then slowly withdraw it away from the coil while the current remains on. The repeatedly reversing field, gradually weakening at the magnet, reduces net domain alignment.

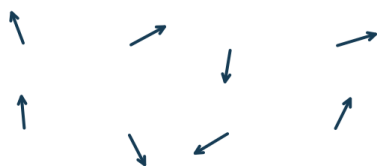
(c) Induction and field diagram (2 marks)

When soft iron is placed near a magnet, the external field aligns its domains. The end nearest a north pole becomes an induced south pole, so attraction occurs. Soft iron loses most of its magnetism when the inducing field is removed.

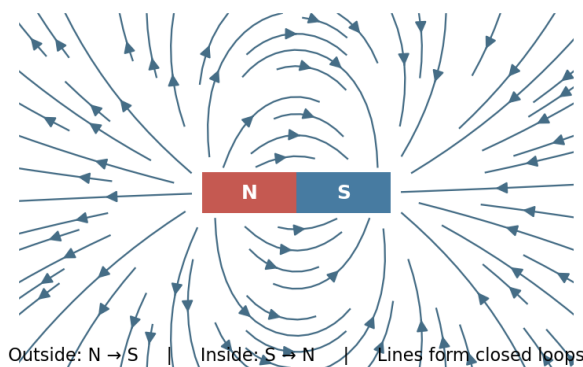
Around a bar magnet, field lines go from **N to S outside** and return **S to N inside**. Lines form closed loops and are closer together near the poles.

Exam care: Do not say that breaking a magnet separates its poles: each piece forms a smaller magnet with both poles.

Unmagnetized: domains point different ways



Magnetized: many domains align



C. LONG-QUESTION SOLUTIONS | L12

L12. Applications of magnets

- (a) Explain how an electromagnet operates an electric bell, including the repeated making and breaking of the circuit. (2 marks)
- (b) Explain how magnetic recording stores information in a magnetizable medium. (2 marks)
- (c) Explain magnetic shielding with soft iron and why magnetic storage media should be protected from strong external magnetic fields. (2 marks)

Theory SLOs: P-09-E-10, P-09-E-14, P-09-E-15

(a) Electric bell (2 marks)

Pressing the switch completes the circuit through a coil on a soft-iron core. The core becomes an electromagnet and attracts an iron armature. The attached hammer strikes the gong. Movement of the armature opens the contact, breaking the circuit. The electromagnet loses its magnetism and a spring returns the armature, restoring contact. This make-and-break cycle repeats while the switch is pressed.

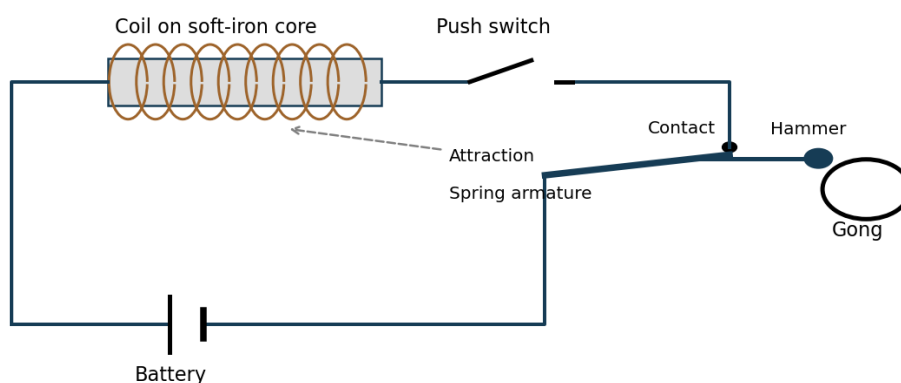
(b) Magnetic recording (2 marks)

A recording head acts as an electromagnet. The varying signal changes the field of the head, magnetizing small regions of a suitable tape or disk in patterns that represent information. The medium retains those patterns after the recording field is removed. A reading head detects changes in magnetization and converts them to an electrical signal.

(c) Shielding and protection (2 marks)

Soft iron has high magnetic permeability. A suitable enclosure provides an easy path for magnetic flux around the protected region, reducing the magnetic field inside. Strong external fields can change recorded magnetization, corrupting information on magnetic tapes, disks or magnetic-stripe cards. Such media should therefore be kept away from strong magnets.

Scope: Magnetic recording applies to magnetizable storage media. Do not claim that all modern electronic memories store information magnetically.



NUMERICAL ANSWER CHECK

Question	Final results
N1	2.5×10^{-8} km; $3.4 \mu\text{s}$; 7.2×10^7 N
N2	10 m, 53.1° north of east
N3	24 m; 8 s; 3.0 m/s
N4	$+4.0 \text{ m/s}^2$; -4.0 m/s^2 ; 300 m
N5	3.0 s; 44.1 m
N6	8.0 kg m/s; 8.0 N s; 40 N
N7	1.6 m/s in the first trolley's original direction
N8	20 N downward; 12 N m each; 50 N upward
N9	300 N/m; 21 N
N10	24.7 m; 20.7 J against resistance
N11	24,500 N; 1633 N; 2.72×10^6 Pa
N12	4.3988×10^7 m; $7331 \text{ m/s} = 7.33 \text{ km/s}$

CHECKLIST FOR EXAM ANSWERS

1. Start with the exact definition or law asked for.
2. Explain the cause and the effect in reasoning questions.
3. Write comparisons as matched points.
4. Label axes, forces, poles, parts and directions in diagrams.
5. Use perpendicular distance in moments; use total distance/total time for average speed.
6. Convert centimetres to metres and cm^2 to m^2 when SI units are needed.
7. State the sign or direction of vector answers.
8. Attempt every requested subpart and retain correct SI units.

Source: FBISE_Physics_9_Theory_Guess_Paper_2nd_Annual_2026.pdf, prepared from the supplied theory summative framework and two subjective sample papers. This is a teacher practice solution, not an official FBISE answer key.